

CG3201 Tutorial 1

Question 1: In one-vs-all multi-class SVM, explain how decision boundaries are determined for a three-class problem with labels $\{A, B, C\}$, and how the prediction is carried out. How many SVM classifiers are required?

Question 2: Given the optimization objective for a soft-margin Support Vector Machine (SVM) as:

$$\min_{\mathbf{w}, b, \boldsymbol{\varepsilon}} \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_i \varepsilon_i$$

subject to: $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \varepsilon_i$, $\varepsilon_i \geq 0, \forall i$, where $\mathbf{w}$ denotes the vector that is perpendicular to the decision hyperplane, b is the offset, $\mathbf{x}_i$ is the i -th training data point, y_i is the label of $\mathbf{x}_i$, ε_i is the slack variable for $\mathbf{x}_i$, and C is the regularization parameter. Its corresponding Lagrangian function is:

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\varepsilon}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{w}, b, \boldsymbol{\varepsilon}) - \sum_i \lambda_i g_i(\mathbf{w}, b, \boldsymbol{\varepsilon}) + \mu_i h_i(\mathbf{w}, b, \boldsymbol{\varepsilon}),$$

where $\lambda_i, \mu_i \geq 0, \forall i$. $f(\mathbf{w}, b, \boldsymbol{\varepsilon}) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_i \varepsilon_i$, $g_i(\mathbf{w}, b, \boldsymbol{\varepsilon}) = y_i(\mathbf{w}^T \mathbf{x}_i + b) + \varepsilon_i - 1$, $h_i(\mathbf{w}, b, \boldsymbol{\varepsilon}) = \varepsilon_i$. Explain, using complementary slackness in the Karush-Kuhn-Tucker (KKT) conditions, why only certain points (misclassified points, points inside the margin, or points on the margin) are support vectors, while other points are not.

Question 3: A binary classification dataset has a curved class boundary, some overlap between the classes, and a few suspected noisy labels. One student proposes a hard-margin linear SVM, while another proposes an RBF-kernel soft-margin SVM with very large values of both C and γ . Compare these proposals and recommend a suitable SVM approach. You should address the following :

- a) Why a hard-margin linear SVM is unsuitable for this dataset;
- b) How slack variables, hinge loss, and C handle overlap and noisy samples;
- c) How the RBF kernel and γ affect the flexibility of the decision boundary;
- d) The overfitting risk of choosing large C and γ , and how these hyperparameters should be selected; and
- e) Which training samples are likely to become support vectors and why the final prediction depends mainly on them.

Question 4: For the soft-margin SVM, determine if the following statements are correct or not correct, and provide detailed justification for your judgement.

- 1) One use of kernel functions in SVM is to decrease the dimensionality of the input features.
- 2) The dimension of the hyperplane depends upon the number of support vectors.
- 3) Deleting the support vectors won't change the position of the hyperplane.
- 4) Soft margin SVM can deal with non-separable datasets, but still can be used together with kernel tricks.

Question 5: For a linearly separable binary dataset trained using a hard-margin SVM, determine if the following statements are correct or not correct, and provide detailed justification for your judgement.

- 1) The distance between the two margin hyperplanes is $\frac{2}{\|w\|}$; thus, minimizing $\frac{1}{2} \|w\|^2$ maximizes the geometric margin.
- 2) Multiplying both w and b by the same positive constant changes the decision boundary and hence changes the predicted class labels.
- 3) Every training sample must satisfy $y_i(w^T x_i + b) \geq 1$.
- 4) If the two classes overlap, a hard-margin SVM remains feasible as long as the regularization parameter C is sufficiently small.

Question 6: For soft-margin SVM, the hinge loss of x_i is $l_i = \max(0, 1 - y_i f(x_i))$. determine if the following statements are correct or not correct and provide detailed justification for your judgement.

- 1) A sample with $y_i f(x_i) = 1.4$ has zero hinge loss.
- 2) A sample with $y_i f(x_i) = 0.4$ has hinge loss 0.4.
- 3) A sample with $y_i f(x_i) = -0.2$ is correctly classified but lies inside the margin.
- 4) Increasing C generally penalizes margin violations more strongly and may lead the classifier to accept a narrower margin to fit more training samples.

Question 7: For a kernel SVM, determine if the following statements are correct or not correct, and provide detailed justification for your judgement.

- 1) For an RBF kernel, increasing gamma gives each training point a wider region of influence and usually produces a smoother decision boundary.
- 2) The kernel trick replaces feature-space inner products with $K(x_i, x_j)$, so the mapping $\phi(x)$ need not be computed explicitly.
- 3) For any finite collection of samples, a valid kernel produces a positive-semidefinite Gram matrix.
- 4) After kernelization, every training sample contributes equally to the decision function, regardless of its optimized dual coefficient.